

```
#include <Wire.h>

#include <MPU6050.h>

#include "BluetoothSerial.h"

#include <TinyGPS++.h>

#include <HardwareSerial.h>


MPU6050 mpu;

BluetoothSerial SerialBT;

TinyGPSPPlus gps;

HardwareSerial GPS_Serial(1); // Use UART1 for GPS


const int buzzerPin = 26;    // Buzzer Pin
const int buttonPin = 12;    // Fall cancel button
const int sosButtonPin = 13; // Manual SOS button
const int IMPACT_THRESHOLD = 20000;


bool fallDetected = false;
unsigned long fallTime = 0;


void setup() {
    Serial.begin(9600);
    SerialBT.begin("ESP32_Fall_Alert");
    Wire.begin();
    mpu.initialize();

    GPS_Serial.begin(9600, SERIAL_8N1, 16, 17); // GPS on UART1


    pinMode(buttonPin, INPUT_PULLUP);
    pinMode(sosButtonPin, INPUT_PULLUP); // SOS button
    pinMode(buzzerPin, OUTPUT);
```

```

if (!mpu.testConnection()) {

    Serial.println(" ❌ MPU6050 connection failed!");

    while (1);

}

Serial.println(" ✅ MPU6050 connected!");

}

void loop() {

    // Check for manual SOS button press
    if (digitalRead(sosButtonPin) == LOW) {

        Serial.println("Manual SOS button pressed!");

        String sosData = "SOS ALERT! URGENT HELP REQUIRED!!:";

        if (gps.location.isValid()) {

            sosData += "Lat=" + String(gps.location.lat(), 6) + ";Lon=" + String(gps.location.lng(), 6);

        } else {

            sosData += "Location Not Available";

        }

        SerialBT.println(sosData);

        Serial.println(sosData);

        delay(2000); // Debounce delay to avoid multiple sends

    }

    // Read acceleration data
    int16_t ax, ay, az;

    mpu.getAcceleration(&ax, &ay, &az);

    long totalAcc = sqrt(ax * ax + ay * ay + az * az);

    Serial.print("Total Acc: ");

```

```
Serial.println(totalAcc);
```

```
// Read GPS Data
```

```
while (GPS_Serial.available()) {  
    gps.encode(GPS_Serial.read());  
}
```

```
if (totalAcc > IMPACT_THRESHOLD) {
```

```
    Serial.println(" 🚨 Sudden movement detected! ALERT!");
```

```
    fallDetected = true;
```

```
    fallTime = millis();
```

```
    digitalWrite(buzzerPin, HIGH);
```

```
    Serial.println(" 📢 Buzzer ON! Press button to cancel alert.");
```

```
    while (millis() - fallTime < 5000) {
```

```
        if (digitalRead(buttonPin) == LOW) {
```

```
            Serial.println(" ✅ Alert canceled by user.");
```

```
            fallDetected = false;
```

```
            digitalWrite(buzzerPin, LOW);
```

```
            return;
```

```
        }
```

```
        delay(500);
```

```
    }
```

```
    Serial.println(" 📶 No response! Sending Bluetooth alert...");
```

```
    String gpsData = "FALL DETECTED! NEED HELP!:";
```

```
    if (gps.location.isValid()) {
```

```
        gpsData += "Lat=" + String(gps.location.lat(), 6) + ";Lon=" + String(gps.location.lng(), 6);
```

```
    } else {
```

```
        gpsData += "Location Not Available";
```

```
}
```

```
SerialBT.println(gpsData);
```

```
Serial.println(gpsData);
```

```
fallDetected = false;
```

```
digitalWrite(buzzerPin, LOW);
```

```
}
```

```
delay(500);
```

```
}
```